Supplementary Material

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

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S1 Data set

Table S1: Data used in this study. All systems: LiTFSI in an amorphous polymer, PCFF+, 353 K.

	5 ns runs	50 ns runs
Trajectories	6,137	133
MSD curves used (Li^+, TFSI)	100 lags each	100 lags each
Longest lag in the curve	5 ns	50 ns
Systems also simulated at the other length	133	133
Systems simulated twice at this length	94	n/a

Raw trajectories of the 133 long runs were read as a stream, keeping only ion positions (lithium and the nitrogen of TFSI), the polymer atoms of the types that the platform lists as solvating, and the system centre of mass. That list of types is the same for every system while the type numbering is not, so the kept polymer atoms are not always those that coordinate lithium (often they are hydrogen or carbon atoms); they are used only to follow the mobility of the polymer. The origin-averaged lithium MSD computed from the extracted ion positions reproduces the curves distributed by the platform to within a relative deviation of 5×10^{-7} in every run, for lags up to 80% of the run. Without subtracting the centre-of-mass motion, the MSD at a lag of 25 ns would be inflated by a median factor of 1.2 and by up to a factor of 34 (more than 10% in 63% of the runs); the largest factors occur in the slowest systems, where a drift of the whole system by a few nanometres is comparable to the displacement of the lithium ions.

Availability and sampling. The folder name `rand_100` indicates a nominal random sample of 100 systems. The released trajectory index contains 95 new-configuration runs: 94 have an original screening run, while system `9-0-489993648-0` is available only as a replica. The index does not explain the 5 other nominal entries, so missingness cannot be assessed fully. Of the 96 screening systems with a long run, 92 also have a replica. These two subsets therefore provide largely overlapping evidence.

The primary population contains one original label per screening system (6,003 systems), excluding literature polymers and the replica-only system. The descriptive audit below compares the replicated systems with the rest. Similar marginal labels or modest descriptor differences do not prove equal noise across chemistry. Connector markers Au and Cu are excluded from heavy atom counts; these markers are retained in the model fingerprints to match the released monomers.

Table S2: Sample audit. The first two numeric columns are medians; the final column is the mean difference (replicated minus rest) divided by the standard deviation in the rest. These are descriptive comparisons, not equivalence tests.

Descriptor	Replicated	Rest	Mean difference / SD
Heavy atoms (excluding connectors)	14.00	14.00	-0.07
O atoms	3.00	3.00	-0.13
N atoms	1.00	1.00	+0.12
F atoms	0.00	0.00	+0.08
Rings	0.00	0.00	–
$\ln D_{\mathrm{end}}$, 5 ns	-1.83	-1.87	+0.05

Table S3: Sensitivity of reliability to the denominator population. The noise estimate uses the same 94 replicated systems in both columns; the broader population additionally includes literature trajectories and repeated labels.

Label	Original screening systems	All short trajectories
End-point	0.44 (0.27 to 0.61)	0.47 (0.31 to 0.62)
Origin-averaged	0.54 (0.40 to 0.67)	0.58 (0.44 to 0.70)

S2 Sensitivity of the regime census and of the lag dependence

Table S4: Sensitivity of the regime census to the lag window used to fit the effective exponent. The main text uses 0.2 to 0.6.

	5 ns runs			50 ns runs		
Fit window (fraction of run)	Median β	$\beta < 0.7$	$\beta \geq 0.9$	Median β	$\beta < 0.7$	$\beta \geq 0.9$
0.1 to 0.5	0.38	99.7%	0.0%	0.60	81.2%	0.0%
0.2 to 0.6	0.43	97.8%	0.03%	0.65	69.2%	3.8%
0.3 to 0.7	0.48	94.9%	0.07%	0.69	54.9%	5.3%
0.4 to 0.8	0.51	87.4%	0.75%	0.73	45.9%	14.3%

Table S5: Sensitivity of the lag dependence in the 133 long runs to the reference lag τ_{ref}. The main text uses 25 ns.

	Median $D_{\mathrm{app}}(\tau)/D_{\mathrm{app}}(\tau_{\mathrm{ref}})$			Spearman ρ		
τ_{ref} (ns)	$\tau = 1$	$\tau = 5$	$\tau = 10$	$\tau = 1$	$\tau = 5$	$\tau = 10$
15	4.43	1.65	1.18	0.95	0.98	0.99
20	4.93	1.83	1.30	0.93	0.97	0.99
25	5.23	1.95	1.40	0.92	0.96	0.98
30	5.56	2.05	1.47	0.92	0.96	0.98
35	5.76	2.14	1.53	0.92	0.96	0.97

Table S6: Apparent diffusivity and ranking as a function of the observed lag in the long runs, relative to a lag of 25 ns (curve-based).

Lag (ns)	Median ratio	10th to 90th percentile	Within 25%	Spearman ρ
0.5	8.86	4.70 to 12.47	0.0%	0.91
1	5.23	3.12 to 7.26	0.0%	0.92
2	3.34	2.15 to 4.32	0.0%	0.94
5	1.95	1.47 to 2.40	0.8%	0.96
10	1.40	1.18 to 1.63	16.5%	0.98
15	1.19	1.08 to 1.31	73.7%	0.99
20	1.07	1.03 to 1.12	100.0%	1.00

S3 Slowdown within the long runs

Each long run is split into ten consecutive 5 ns segments, and the first segment is compared with the five late ones (25 to 50 ns). Mobilities are end-point estimates over each segment, after subtracting the centre-of-mass motion, for lithium, for the nitrogen of TFSI and for the polymer atoms kept from the raw trajectory (125 systems that have any). Production is at constant volume (NVT); the potential energy and pressure are the segment means of the 100 ps averages written by the simulations. Ion clustering uses the lithium-O(TFSI) clusters computed in every frame during the simulations: the size of the cluster that contains each lithium, and the fraction of lithium ions whose cluster contains no anion oxygen; the number of anion nitrogens around each lithium uses a cutoff at the first minimum of the lithium-nitrogen radial distribution function of each system.

Table S7: What changes between the first 5 ns segment of each long run and the late segments (133 runs). Potential energy is in kcal mol^{-1} per atom. Mobilities are ratios first/late (median and bootstrap 95% interval); the other quantities are differences first − late. "Late segments" gives the direction of the change, "Systems" the share of runs in which it occurs, and ρ the Spearman correlation, across systems, with the slowdown of lithium.

Changes across all systems

Quantity	First vs late (95% CI)	Late	Systems
Lithium diffusivity (ratio)	1.63 (1.52 to 1.81)	slower	93%
Anion diffusivity (ratio)	1.46 (1.39 to 1.61)	slower	89%
Polymer-atom mobility (ratio)	1.43 (1.37 to 1.51)	slower	96%
Potential energy per atom	0.014 (0.013 to 0.015)	lower	100%
Pressure (atm)	69 (50 to 79)	lower	83%
Size of the cluster of a Li (ions)	0.37 (0.32 to 0.44)	smaller	85%
Fraction of Li without anion	−0.03 (−0.04 to −0.02)	larger	75%
Anion N around each Li	0.06 (0.04 to 0.08)	fewer	82%

Subsets and associations

Quantity	Screening	Literature	ρ with Li (95% CI)
Lithium diffusivity (ratio)	1.80	1.28	–
Anion diffusivity (ratio)	1.57	1.32	0.70 (0.60 to 0.77)
Polymer-atom mobility (ratio)	1.50	1.26	0.68 (0.57 to 0.76)
Potential energy per atom	0.015	0.011	0.58 (0.46 to 0.68)
Pressure (atm)	79	34	0.30 (0.13 to 0.45)
Size of the cluster of a Li (ions)	0.37	0.37	−0.03 (−0.20 to 0.15)
Fraction of Li without anion	−0.02	−0.05	0.15 (−0.02 to 0.31)
Anion N around each Li	0.07	0.04	−0.06 (−0.24 to 0.10)

S4 Run-to-run reproducibility

Table S8: Reproducibility of a single 5 ns run, from 94 systems simulated twice from different initial structures. Intervals by bootstrap over systems.

Estimator	σ_{run}	Factor	Rank corr.	r (95% CI)	ICC (95% CI)
End-point	0.35	1.43	0.48	0.44 (0.27 to 0.61)	0.53 (0.34 to 0.66)
Origin-averaged, lag $T/2$	0.28	1.32	0.56	0.54 (0.40 to 0.67)	0.62 (0.45 to 0.74)

Two runs of the same system differ by more than a factor of 1.5 in 38% of the replicated systems. The original-label spread in the replicated subset and the primary screening population is reported in the sample audit. The reliability denominator uses one original label per screening system; the broader set of all short trajectories is a sensitivity analysis (below). The effective exponent of an individual 5 ns run is not reproducible (noise 0.15, rank correlation between runs -0.07): it characterises the population, not a single system.

Structural and dynamic noise. The dynamic noise is estimated from the 96 screening pairs (the 5 ns run against the first 5 ns of the long run from the same structure) and the total from the 94 replicated systems. The short run of a pair is the original replica of the same system, so the two differences are correlated (correlation 0.53 on the 92 systems that have both). Resampling those systems jointly, the structure accounts for 57% of the run-to-run variance of the end-point label (44 to 69%; separate resampling of the two sets, as in the main text: 55%, 28 to 73%) and 52% (33 to 66%) of that of the origin-averaged label.

Batches. In the data set the original screening runs, the replicas and the long runs belong to three different batches. Constant shifts or changes in marginal spread are possible batch effects; system-specific effects need not produce either. Neither marginal difference is detected for the end-point label: the mean ratio original to replica is 1.05 (95% interval 0.95 to 1.17, $p = 0.31$), the mean ratio of the short run to the first 5 ns of its long run is 1.00 (0.93 to 1.06, $p = 0.89$), and the spread of $\ln D_{\mathrm{app}}$ across systems is 0.50 for the originals and 0.53 for their replicas (Levene $p = 0.67$; 0.47 in the rest of the screening set). The noise estimates use the standard deviation of the differences, which excludes a constant offset; without removing it the reliability is 0.44, unchanged. Batch and initial structure cannot be fully separated, because every replica is both new and from a later batch.

S5 Ion aggregation: exploratory associations

The platform's cluster populations provide the fraction of lithium without an anion and in clusters containing at least five lithium ions. We relate these descriptors to the long-run reference, short-run overestimation and ranking error. The latter two are also examined using partial Spearman correlations controlling for $\ln D_{50}$, because mobility and the bias measure share the long-run reference. All intervals are system bootstraps; these exploratory comparisons are not adjusted for multiple testing and do not establish a mechanism.

Polymer-contact descriptors from the initial extraction are not interpreted: the platform's fixed list of polymer atom types does not identify the same chemical species in every system. In particular, contacts with those retained atoms cannot be treated as a coordination shell or used to explain the structural noise component. Their cached values remain archived for transparency but are excluded from the chemical conclusions.

Table S9: Exploratory aggregation associations (133 long runs). Entries are Spearman correlations with 95% bootstrap intervals; partial correlations control for long-run mobility. Intervals spanning zero limit mechanistic interpretation.

Descriptor	Outcome	Raw correlation	Partial correlation
Free Li	$\ln D_{50}$	-0.02 (-0.20, +0.17)	–
Free Li	Overestimation	-0.17 (-0.35, +0.01)	-0.20 (-0.35, -0.03)
Free Li	Rank error	-0.06 (-0.23, +0.11)	-0.07 (-0.22, +0.10)
Li in large clusters	$\ln D_{50}$	-0.25 (-0.41, -0.08)	–
Li in large clusters	Overestimation	+0.26 (+0.10, +0.42)	+0.18 (-0.00, +0.34)
Li in large clusters	Rank error	+0.18 (+0.02, +0.34)	+0.10 (-0.06, +0.26)

S6 New runs with the original protocol

A SCAYLE pilot selected 10 screening systems at random (seed 20261104) among the systems with an original run, a replica and a long run. The planned five runs per system comprised one production re-run from the original structure and velocities, two with new velocities from the same structure, and two from annealed structures. The production settings were NVT at 353 K, PCFF+, and 5 ns. Annealing used 600 K, cooling, and NPT re-equilibration; it is a different preparation from the dataset's original generator. Of 50 planned runs, 49 were completed in the supplied snapshot.

The geometric-mean re-run/original ratio is 1.05 (95% interval 0.92 to 1.21). The estimated within-system standard deviation across the available original and same-structure production labels is 0.23 (0.19 to 0.27), compatible with 0.24 from the screening-pair comparison. Shared initial histories and the small sample limit this check; it is not evidence of complete dynamical independence. The available pairs of annealed runs give dispersion 0.20 (0.10 to 0.28), compared with 0.23 (0.10 to 0.34) for the dataset's original/replica pairs in the same systems. These small-sample estimates do not resolve a structural variance component. The annealed/original geometric-mean ratio is 0.86 (0.70 to 1.04), whose interval includes one; a systematic shift is not established.

The public compact-input release contains the authors' aggregate summary and its provenance. The intervals were produced by system bootstrap in the private pilot analysis; per-run labels are not included in this snapshot. The release regenerates the reported macros from those summaries, rather than recalculating pilot bootstrap intervals. This pilot checks the protocol and noise scale, not the efficacy of adaptive allocation.

S7 Predictor split checks

Chemical-family cross-validation uses Butina clustering with Tanimoto similarity thresholds 0.6 (1,397 groups) and 0.4 (130 groups). The median similarity to the nearest training monomer is 0.67 and 0.65, respectively, against 0.72 for the system split. Cluster membership does not guarantee a pairwise train–test similarity cap, so these splits do not establish generalization to distant chemistry.

In the trajectory split, 77 of the 94 replica pairs fall in different folds. On their 154 labels, predictor correlation is 0.66 for trajectory splitting and 0.67 for system splitting. This small difference is specific to the model and data; keeping all trajectories of each system together remains necessary to avoid leakage.

S8 End-point estimator

The end-point value of the 5 ns runs deviates from the power law fitted over 20% to 60% of the same curve with a standard deviation of 0.19 in log units, a typical factor of 1.21 (Fig. S1); 5.4% of the systems deviate by more than a factor of 1.5, and the rank correlation between the end-point estimate and the origin-averaged estimate at half the run length is 0.92. This deviation combines the single-origin noise of the end point with any departure of the curve from a power law; the replicated runs (Table S8) give the direct comparison of the two estimators.

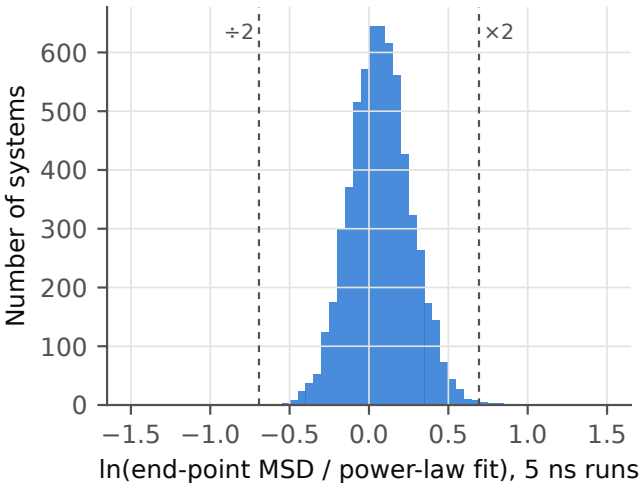


Figure S1: Deviation of the end-point MSD of the 5 ns runs from the power law fitted over lags between 20% and 60% of the run. Dashed lines mark a factor of two.

S9 Paired short and long runs

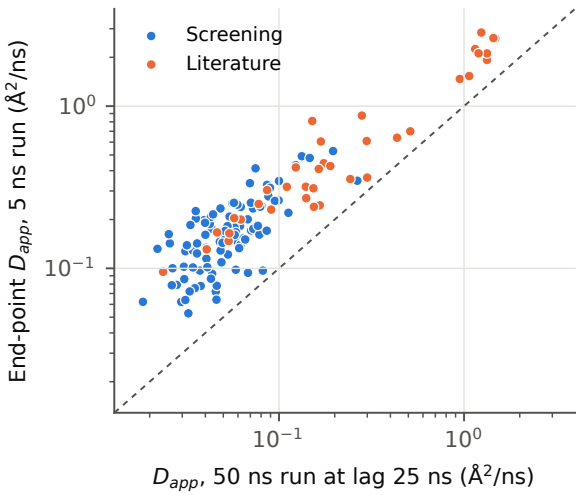


Figure S2: End-point apparent diffusivity from the 5 ns run against the origin-averaged apparent diffusivity at a lag of 25 ns from the 50 ns run of the same system, which starts from the same initial structure (133 systems). The dashed line is the identity.

S10 Prefix-based estimators and stopping rules

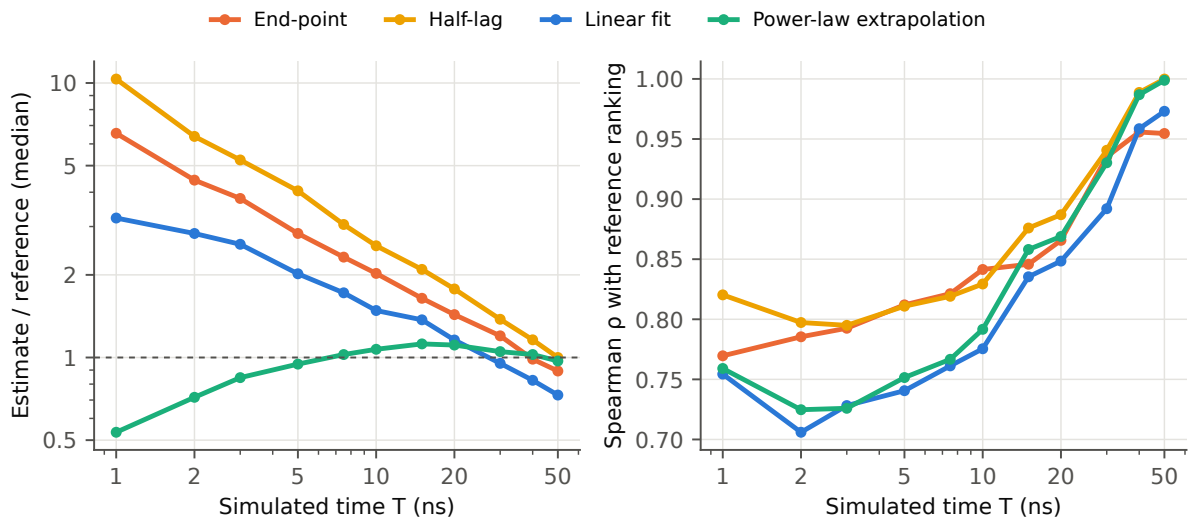


Figure S3: Estimates from prefixes of the long trajectories. Left: median ratio to the finite-lag reference $D_{\mathrm{app}}(25\,\mathrm{ns})$; right: ranking correlation with that reference. Reference and prefixes share trajectory data.

Table S10: Estimates from a prefix of length T relative to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ of the full 50 ns run: median ratio and typical multiplicative error (133 runs).

	End-point		Half-lag		Linear fit		Power-law extrapolation	
T (ns)	Ratio	Error	Ratio	Error	Ratio	Error	Ratio	Error
1	6.56	×6.56	10.34	×10.34	3.22	×3.27	0.53	×1.87
2	4.43	×4.43	6.39	×6.39	2.83	×2.85	0.72	×1.50
5	2.83	×2.83	4.05	×4.05	2.02	×2.02	0.95	×1.37
10	2.02	×2.02	2.55	×2.55	1.48	×1.51	1.07	×1.26
20	1.43	×1.45	1.78	×1.78	1.16	×1.24	1.11	×1.20
30	1.20	×1.23	1.38	×1.38	0.95	×1.20	1.05	×1.13
50	0.89	×1.13	1.00	×1.00	0.73	×1.37	0.97	×1.03

Table S11: Sensitivity of the prefix-based comparison to the reference (133 long runs). Each cell gives the typical multiplicative error and, in parentheses, the median ratio of the estimate to the reference. The power law is extrapolated to the quantity that defines each reference: the apparent diffusivity at that lag, or the mean slope of $A\tau^{\beta}$ over 20 to 40 ns.

Reference (from the 50 ns run)	End-point, 5 ns	Linear fit, 5 ns	Power law, 5 ns	Power law, 10 ns
D_{app} at 15 ns	×2.37 (2.37)	×1.80 (1.73)	×1.33 (1.06)	×1.22 (1.16)
D_{app} at 25 ns (main)	×2.83 (2.83)	×2.02 (2.02)	×1.37 (0.95)	×1.26 (1.07)
D_{app} at 35 ns	×3.15 (3.15)	×2.23 (2.23)	×1.39 (0.86)	×1.32 (1.00)
D_{app} at 40 ns	×3.27 (3.27)	×2.33 (2.33)	×1.45 (0.83)	×1.30 (0.95)
Slope of MSD, 20 to 40 ns	×4.01 (4.01)	×3.05 (3.05)	×1.99 (0.54)	×1.52 (0.68)

The regime rules rarely fire: the effective exponent of a prefix reaches 0.9 before 50 ns in 3%

Table S12: Stopping rules evaluated on the long runs (133 runs). Runs that do not meet a rule before 50 ns are simulated for the full length and evaluated there. Mean time and the first error column are averaged over all runs; the last column restricts the error to the runs that stop early.

Rule	Estimator	Stop before 50 ns	Mean time (ns)	Error, all runs	Error, runs that stop
Regime: $\beta \geq 0.9$	Linear fit	3%	49.3	×1.37	×1.38
Regime: $\beta \geq 0.8$	Linear fit	19%	44.6	×1.39	×1.39
Precision: CV < 10%	Linear fit	2%	49.3	×1.37	×1.36
Precision: CV < 20%	Linear fit	80%	15.6	×1.79	×2.22
Precision: CV < 10%	Power law	24%	43.4	×1.04	×1.13
Fixed 5 ns	End-point	n/a	5.0	×2.83	n/a
Fixed 5 ns	Linear fit	n/a	5.0	×2.02	n/a
Fixed 10 ns	Linear fit	n/a	10.0	×1.51	n/a
Fixed 20 ns	Linear fit	n/a	20.0	×1.24	n/a
Fixed 5 ns	Power law	n/a	5.0	×1.37	n/a
Fixed 10 ns	Power law	n/a	10.0	×1.26	n/a
Fixed 20 ns	Power law	n/a	20.0	×1.20	n/a

of the runs and 0.8 in 19%, so they save almost no simulation, and when the $\beta \geq 0.8$ rule fires the error of the runs that stop is 1.39. The precision rule with a 20% tolerance stops early in 80% of the runs but with a typical error of 2.22 for those runs: a small bootstrap CV measures the precision of a lag-dependent quantity, not its distance from the long-lag value.

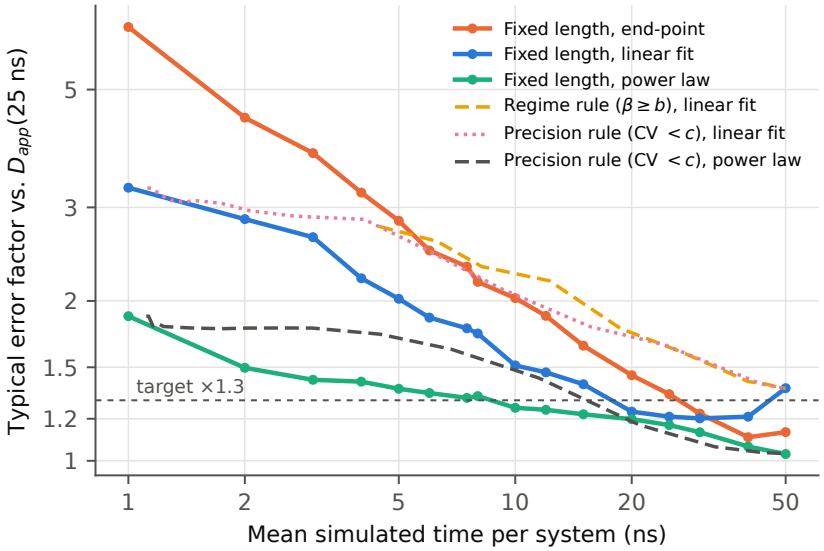


Figure S4: Cost-error frontier of estimators and stopping rules on the 133 long runs. Each line follows one rule as its parameter varies; runs that do not stop before 50 ns count as 50 ns. Fixed-length power-law extrapolation forms the lower envelope: a typical error of ×1.5 requires 2.0 ns with it, against 12.0 ns with a linear fit and 20.0 ns with the end-point estimate (10.0, 20.0 and 30.0 ns for ×1.3).

Table S13: Settings chosen on half of the systems and evaluated on the other half (median over 500 random partitions): mean simulated time, typical error, and percentage of partitions in which the held-out half meets the target; n/a means that no setting met the target on the half used for the choice.

	Target error ×1.5			Target error ×1.3		
Rule	Time (ns)	Error	Target met	Time (ns)	Error	Target met
Fixed length, end-point	20.0	×1.45	97%	30.0	×1.23	84%
Fixed length, linear fit	12.0	×1.47	61%	20.0	×1.24	96%
Fixed length, power law	2.0	×1.50	59%	7.5	×1.31	45%
Regime rule, linear fit	35.2	×1.48	64%	n/a	n/a	0%
Precision rule, linear fit	37.0	×1.47	67%	50.0	×1.40	0%
Precision rule, power law	11.9	×1.43	76%	20.9	×1.18	93%

S11 Adaptive allocation: algorithm and robustness

Algorithm. All quantities are in $\ln D$ of the end-point label. A campaign has $N = 6{,}003$ candidates, a target of the top k and a mean budget of b runs per candidate, each run starting from a new, equilibrated structure.

1. *Model.* System values $\mu_i \sim \mathcal{N}(0, \sigma_{\text{sys}}^2)$ with $\sigma_{\text{sys}} = 0.31$; a run of system i is $\mu_i + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma(\mu_i)^2)$ and $\sigma(\mu) = \max\{a + \gamma\,\mu/\sigma_{\text{sys}}, 0.08\}$, $a = 0.35$, $\gamma = -0.067$, fitted so that the mean noise of the slower and faster halves matches the replicated systems.
2. *Posterior.* After n_i runs with mean $\bar{y}_i$, the posterior of μ_i is approximated by a normal with precision $1/\sigma_{\text{sys}}^2 + n_i/\sigma(\bar{y}_i)^2$ and mean $(n_i\bar{y}_i/\sigma(\bar{y}_i)^2)$ divided by that precision. Because the noise depends on μ_i, the exact posterior is not normal; the approximation evaluates the noise at the observed mean (plug-in). Allocating instead with a constant-noise model, whose posterior is exactly normal, gives practically the same recall while the runs keep their mobility-dependent noise (see below), so the approximation does not drive the results.
3. *First round.* One run for every candidate.
4. *Batches.* While budget remains: take the cut-off c_k as the k-th largest posterior mean, compute $p_i = \Phi[(m_i - c_k)/s_i]$ from the posterior mean m_i and standard deviation s_i, and give one more run to the 5% of candidates with the largest $p_i(1 - p_i)$.
5. *Selection.* The top k by posterior mean. The metric is the fraction of the true top k, the k largest μ_i, that is selected.

The true top set is therefore defined by the expected 5 ns label of each system, not by its long-time diffusivity. Uniform replicates, successive halving and the two-stage re-screening use the same model, posterior and selection; only the choice of who receives each run differs. All random draws use fixed seeds, given in the code.

The saving for the top 10% against four uniform replicates is ×2.1 when the system values follow the empirical shape of the original screening labels instead of a normal distribution, and ×2.3 with the origin-averaged label. Using the mobility-dependent noise in the allocation does not help: the recall of the top 10% with four runs per candidate is 82% with it and 84% with a constant noise. Successive halving with restarted rounds performs similarly to boundary-focused allocation. With the end-point label and 1.25 runs per candidate it recovers 71% of the top 1%, against 64%; with the origin-averaged label and four runs per candidate it recovers 81%, against 89%.

Lower planned reliability spreads replicates more widely, but planning at either confidence limit changes savings little (Table S16). When the true reliability instead lies at those limits,

Table S14: Recall of the true top set for each allocation and mean budget (end-point label, Gaussian system values, 200 simulated campaigns of 6,003 candidates); the values plotted in the allocation figure of the main text. Each run starts from a new, equilibrated structure. With one run per candidate the recall of the top 10% is 45%, not the 43% of the noise-ceiling table, because here the noise depends on mobility.

		Mean runs per candidate					
Target	Allocation	1	1.5	2	3	4	6
Top 1%	Uniform replicates	27%	n/a	48%	59%	65%	72%
	Successive halving	27%	79%	83%	87%	89%	89%
	Boundary-focused	27%	71%	79%	84%	86%	90%
Top 5%	Uniform replicates	38%	n/a	55%	63%	68%	74%
	Successive halving	38%	67%	75%	81%	84%	87%
	Boundary-focused	38%	67%	74%	80%	84%	87%
Top 10%	Uniform replicates	45%	n/a	59%	66%	70%	76%
	Successive halving	45%	63%	71%	78%	82%	85%
	Boundary-focused	45%	66%	72%	78%	82%	85%

Table S15: Mean number of 5 ns runs per candidate that boundary-focused allocation needs to match the recall of uniform replicates, and the corresponding saving (end-point label, Gaussian system values, 200 simulated campaigns of 6,003 candidates).

	Uniform replicates matched		
Target	2 runs	3 runs	4 runs
Top 1%	1.1 (×1.7)	1.2 (×2.5)	1.3 (×3.1)
Top 5%	1.2 (×1.7)	1.4 (×2.2)	1.6 (×2.5)
Top 10%	1.2 (×1.6)	1.6 (×1.9)	1.9 (×2.1)

Table S16: Recall at mean budgets of two/four runs (left/right), with saving against four uniform replicates in parentheses. The two-stage column chooses the best shortlist ($2k$, $4k$, $8k$) in hindsight. End-point labels, Gaussian system values, 200 campaigns; Monte Carlo standard error at most 0.5 percentage points (median 0.1).

	Boundary-focused, planned with r =			Two-stage
Target	0.44 (true)	0.27	0.61	re-screening
Top 1%	79% / 86% (×3.1)	76% / 83% (×3.1)	79% / 88% (×3.1)	77% / 82% (×3.2)
Top 5%	74% / 84% (×2.5)	72% / 81% (×2.4)	73% / 83% (×2.5)	71% / 78% (×2.3)
Top 10%	72% / 82% (×2.1)	71% / 81% (×2.1)	71% / 81% (×2.1)	69% / 79% (×1.8)

keeping total label spread fixed and using the correct calibration, savings are ×2.7 to ×3.2 for the top 1%, ×1.7 to ×2.9 for the top 5%, and ×1.6 to ×2.5 for the top 10%. These are model scenarios, not a full uncertainty interval for savings. Two-stage re-screening is sensitive to shortlist size; $8k$ is best for the top 1% in these scenarios. Candidates ranked low initially may receive no further information if the shortlist is too short.

S12 Check of the adaptive allocation on real runs

Table S17: Check of the adaptive allocation on the replicated systems. Calibration: 94 systems, predictions in both directions (first run → second and second → first), noise parameters estimated leaving out the system evaluated. Campaign: 92 systems with a long run; every system receives its first run and 46 receive a second run, chosen at the boundary or at random (mean of 2000 draws); recall of the top set according to the long run. Interval: bootstrap over systems, refitting the noise and rerunning the allocation at a fixed budget of 46 second runs. The random control averages 50 placements per resample.

	Top 10%	Top 20%
Brier improvement over the base rate	35%	16%
Calibration slope (95% CI)	1.00 (0.57 to 2.26)	0.56 (0.29 to 1.31)
Systems in the top set, k	9	18
Recall, first runs only	67%	72%
Recall, second runs at random	71%	70%
Recall, second runs at the boundary	78%	72%
Recall, second runs for all	78%	72%
Boundary − random (95% CI, points)	−12 to +16	−7 to +12

The long runs share the structure of the first run, which can favour it relative to a second initialization. The direction and size of the effect on policy comparisons are not established; it does not explain the top-20% result by itself. Calibration predicts a second noisy label, not latent top-set membership. The campaign also fits noise to its own pairs and is exploratory.